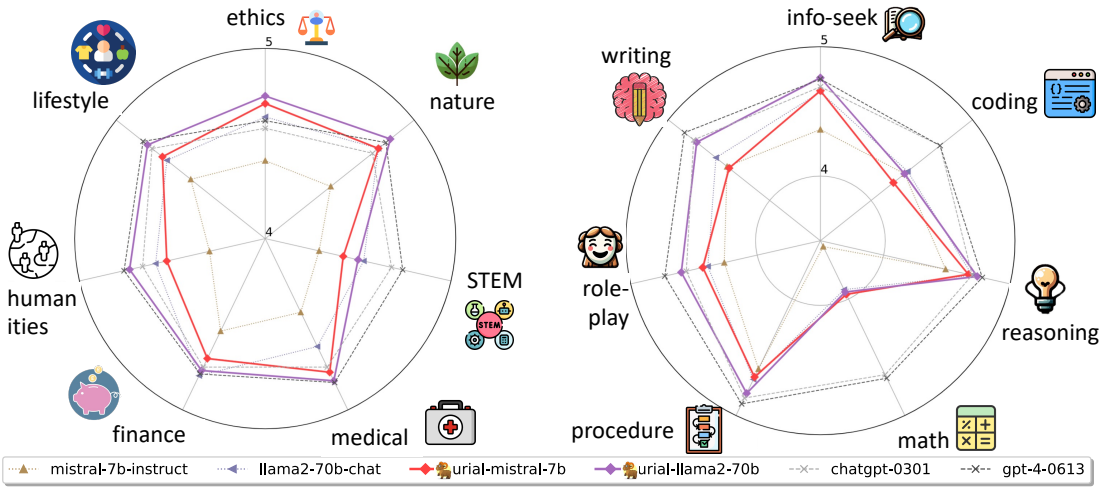

Figure 7: **Categorized performance analysis for alignment in different tasks and topics.**

least in the scenarios covered by our evaluation. We also conduct a human evaluation for pairwise comparisons in Table 2, which reinforces these conclusions.

Aligned LLMs might forget knowledge and become overly sensitive. Case studies in Appendix B reveal that fine-tuning could induce forgetting, hallucination, and overly sensitive censorship. For example, we find that Mistral-7B with URIAL can correctly answer the question "*Did Facebook corporate change its name?*" by telling users the new name is "Meta Platform Inc.". However, the SFT-ed version Mistral-7B-Instruct instead answers "No, Facebook did not change its name." This indicates that during SFT process, the LLMs might over-fit the instruction-tuning data and thus forget the previously acquired knowledge. Also, Llama-2-70b-chat (RLHF-ed) refuses to answer harmless questions such as "*What would be a great twist for a murder mystery? I'm looking for something creative, not to rehash old tropes.*" because it wants to avoid generating any harmful content. In contrast, URIAL can maintain a flexible balance between helpfulness and harmlessness.

What if we use fewer or more in-context examples for URIAL? In the previous analysis, we primarily discussed the performance of URIAL with K=3 in-context examples. To test K=1, we retained only the 'renewable energy' example from Fig. 5. For K=8, we added one example for each of the following topics: mathematics, coding, poetry writing, procedure, and safety. URIAL with K=1 and Llama-2-70b exhibits satisfactory overall performance (4.62), achieving a higher helpfulness score (4.60) than the RLHFed model. With K=8 examples, totaling approximately 2k tokens, we observe a significant improvement in safety for Llama-2-7b (4.29 to 4.97); however, a decrease in performance is seen for all other aspects. On the Mistral-7B, URIAL with K=8 also achieves higher safety and engagement scores. Although URIAL with K=8 demonstrates comparable or better overall performance than URIAL with K=3 for Llama-2-7b, we recommend the use of URIAL with K=3 due to its balanced performance and lower cost.

Is URIAL sensitive to the selection of in-context examples? We also test URIAL with a different set of three examples using Mistral-7b. We find that the overall performance is even slightly better than the default version (4.63 vs. 4.64). The three different examples are available on our website, and the resulting performance metrics are as follows: helpfulness = 4.60; clarity = 4.90; factuality = 4.50; depth = 4.20; engagement = 4.74; safety = 4.93; Average = 4.64; Length = 185.9. This suggests that URIAL is robust to variations in the ICL examples provided.

Can URIAL handle multi-turn conversations? Yes! One can simply append the previous chat history as new in-context examples for URIAL to facilitate multi-turn chatting. We include a case study on using URIAL for multi-turn conversations in Appendix B.3. This suggests that the conversation ability of aligned LLMs might be largely from the base models themselves.

4.4 More insights from Evaluation with Just-Eval-Instruct

Unbiased evaluation. In the first group of results in Table 1, we can see that SFT and RLHF have their own advantages. Llama-2-7b-chat is much safer and more engaging than Vicuna-7b but significantly worse in other aspects such as ℹ helpfulness. Also, the factuality score of Llama-2-7b-chat (RLHFed) also lower than only Vicuna-7b (SFTed), suggesting that RLHF might even cause more hallucination. We conjecture that RLHFed Llama-2-Chat models may have been overfitted to safety and engagement. We also find that Llama-2-chat generates longest outputs that are nearly 250 words on average, while the others are around 150-180 words.

Many prior evaluation (Wang et al., 2023) and leaderboards (Li et al., 2023a) are solely based on *win-rates* on *overall* quality, and thus tend to prefer longer outputs and some particular aspects for unclear reasons. Now with `just-eval-instruct` and our multi-aspect evaluation, we can clearly analyze which aspects is a particular model or alignment method indeed improving. In addition, our evaluation also provides rationales that humans can easily verify.

Gap between open-source LLMs and ChatGPTs. Prior evaluation such as AlpacaEval does not have tags for each example for testing, so it is hard to do large-scale detailed analysis. Open-source LLMs still have gaps to OpenAI GPTs on several tasks and topics. We use Figure 7 to analyze the categorized performance. It is obvious that GPTs have a much more balanced performance on almost all tasks and topics. Open-source LLMs including URIAL are weak on coding and math tasks as well as STEM topics, although they can match the performance of GPTs on other data categories. Please refer to Figure 1 for a visualization of the performance in different aspects.

5 Related Work & Discussion

5.1 Alignment Tuning & Superficial Alignment Hypothesis

Limitation of Tuning-based Alignment. Alignment tuning through SFT and RLHF typically demands substantial resources, such as GPU nodes, a large amount of instruction data, and human annotations, making the process both costly and time-consuming. This restricts ordinary labs from aligning extreme-scale LLMs exceeding 30B, let alone the recent Falcon-180B (Almazrouei et al., 2023). Moreover, during the pre-training and continual training stages, efficiently estimating the downstream performance of a base model checkpoint becomes challenging if alignment tuning is always required to evaluate its instruction-following ability. Besides the aforementioned limitations, tuning-based alignment may also cause forgetting issues in LLMs.

Wang et al. (2023) demonstrated that some SFTed LLMs perform significantly worse than their base counterparts on factual and reasoning benchmarks. For instance, applying SFT to Llama-13b with `self-instruct` (Wang et al., 2022a) results in a considerable decline in its MMLU performance (from 42.5 to 30.3) and Codex-Eval performance (from 26.6 to 13.4). Even more strikingly, SFT with SuperNI (Wang et al., 2022b) causes Llama-13B to nearly lose all its BBH reasoning ability (decreasing from 36.9 to 2.8). Moreover, Shen et al. (2023) show that the reward models in RLHF can perform very inconsistently, yielding a nearly random performance when showing contrastive instructions to them. These findings imply that alignment tuning may lead to the forgetting of previously acquired knowledge in base LLMs, which is also shown in our experiments.

Superficial alignment hypothesis. LIMA (Zhou et al., 2023) employs only 1k examples to fine-tune a 65B LLM and discovers that such a slightly tuned LLM surprisingly achieves a high win rate over ChatGPT, thus implying that the alignment tuning is superficial. Similar observations are also reported by other recent studies (Chen et al., 2023a; Lee et al., 2023). Moreover, Gudibande et al. (2023) demonstrate that aligning open-source LLMs by imitating proprietary LLMs (e.g., ChatGPT) may not always yield desirable results, emphasizing the importance of a strong pre-trained base LLM for producing factual content. Tuning-based methods such as LIMA still require tuning the weights of LLMs and consequently face the limitations described above when the model size is too large or we need to frequently align base LLMs for evaluation. A concurrent work (Duan et al., 2023) also explores the similarity between ICL and instruction-tuning in their effects on downstream tasks by analyzing the LLMs' hidden states. As for the theory of alignment, these studies only indirectly suggest the promise of the superficial alignment hypothesis but do not directly show where and when the alignment tuning significantly changes the model behavior. In this paper, we study the

superficial alignment hypothesis more directly through the lens of token distribution shift, which directly exhibits the alignment effect and produces more detailed non-trivial findings.

5.2 TUNING-FREE ALIGNMENT METHODS

In-context learning. Many in-context learning (ICL) studies focus on specific NLP tasks, such as classification and multiple-choice QA (Wei et al., 2023; Zhang et al., 2022). However, few investigations concentrate on aligning base LLMs for building assistants via in-context learning. Bai et al. (2022) uses a prompt with 14 examples (about 7k tokens) with context distillation for their iterated RLHF training process. In a recent work, Han (2023) demonstrates that in-context learning using approximately 10 *dynamic*, *retrieved* examples can achieve impressive performance in aligning base LLMs, sharing a similar motivation with ReCross (Lin et al., 2022). We treat this as a baseline method (Retrieval ICL in Table 1) and improve it by incorporating more high-quality data and employing state-of-the-art sentence embedding to index and query. Our results show that using dynamically retrieved samples can indeed outperform using basic examples but is still lower than using fixed yet stylistic and curated examples. Ye et al. (2023a) proposes in-context instruction learning that also suggests the promise of using carefully crafted in-context examples can improve the generalization ability of base LLMs. Furthermore, Min et al. (2022) find that ICL primarily concerns the style and format of demonstrations, rather than their truth content, which aligns with our motivation for using curated and stylistic examples.

The **limitations** of ICL for tuning-free alignment are twofold: context length and computational cost. Recent research has significantly increased the context window sizes for LLMs (Su et al., 2021; Tworkowski et al., 2023; Xiong et al., 2023; Chen et al., 2023b), ensuring that URIAL with only 1k tokens does not hinder any long-context applications of LLMs. To reduce computational costs, we can precompute the activations of static prefixes in ICL methods like URIAL and then load them into the KV cache to avoid recomputation for each new inference request. Efficiency can be further enhanced by using advanced caching methods (Ge et al., 2023; Mu et al., 2023; Gim et al., 2023) and optimized attention modules during decoding (Dao et al., 2022; Dao, 2023).

RAIN (Li et al., 2023b) is a recent tuning-free alignment method that shares similar motivation with ours, while they focus on the inference-time self-evaluation for better decoding. In contrast, URIAL only relies on in-context learning and does not have special requirements on decoding strategy. URIAL and RAIN can be potentially integrated with each other for even better inference-time alignment methods that do not require parameter tuning at all.

5.3 EVALUATION OF ALIGNMENT

Previous evaluation approaches, as employed in AlpacaEval (Li et al., 2023a) and MT-bench (Zheng et al., 2023), focus on rating or comparisons with respect to overall performance, which obscure our understanding of the specific strengths and weaknesses of each output candidate. Recent works, such as FLASK (Ye et al., 2023b) and TIGERScore (Jiang et al., 2023b), share similar motivations with ours by utilizing multiple aspects and explanations for evaluating LLMs, though they have different focal points. Our `just-eval-instruct` consolidates 9 existing datasets for testing alignment and provides multi-faceted, explainable evaluations to analyze alignment performance in LLMs, which can be used as an complementary evaluation of LLMs.

5.4 MORE DISCUSSION ON URIAL

Although URIAL can match the performance of SFT and RLHF when the base LLMs are strong, it is not suggested to replace SFT or RLHF with URIAL in all scenarios. Specifically, model tuning may still be necessary for tasks such as coding (Luo et al., 2023), mathematics (Yue et al., 2023), interactive agents (Yin et al., 2023), etc. The unique advantages of URIAL include:

- URIAL is **a strong baseline method** for aligning base LLMs without tuning. It is extremely simple to implement and perfectly reproducible, thus facilitating the development and evaluation of future tuning-free and tuning-based alignment methods.
- URIAL can **align extremely large LMs** (e.g., Llama-2-70b, Falcon-180b) with minimal effort. Fine-tuning such extremely large models requires significant computation resources and time; URIAL aligns them without tuning, thereby saving both.